

Demystified Carathéodory extension theorem

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Carathéodory's extension theorem concerns the striking fact that σ -additivity on an algebra is already enough to extend a set function to a measure on a σ -algebra containing that algebra.

The theorem, due to the Greek mathematician Constantin Carathéodory, gives a general construction of such an extension. But even Kallenberg, not known for rhetorical excess, calls the key step an “ingenious technical result.” The purpose of this note is to arrive at the key step naturally.

Let μ be a σ -additive non-negative set function on an algebra $\mathcal{A}$ of subsets of S , with $\mu(\emptyset) = 0$ and values in $[0, \infty]$. That is, μ is a *premeasure*. Let μ^* be the induced outer measure,

$$\mu^*(T) = \inf \left\{ \sum_{n=1}^{\infty} \mu(A_n) : T \subseteq \bigcup_{n=1}^{\infty} A_n, A_n \in \mathcal{A} \right\}.$$

The induced outer measure agrees with μ on $\mathcal{A}$:

$$\mu^*(A) = \mu(A), \quad A \in \mathcal{A}.$$

Suppose that we want to adjoin a set $E \subseteq S$ to the algebra. Write

$$\mathcal{A}[E] = \text{alg}(\mathcal{A} \cup \{E\}).$$

The outer measure μ^* is already defined on every subset of S . It is therefore natural to ask whether, on the enlarged algebra $\mathcal{A}[E]$, it is *still* a premeasure extending μ .

Definition 1. A set $E \subseteq S$ is *Carathéodory measurable* if

$$\mu^*|_{\mathcal{A}[E]}$$

is a premeasure.

Since $\mu^* = \mu$ on $\mathcal{A}$, such a premeasure automatically extends the one with which we started.

We now recover Carathéodory's classical criterion. We first need the corresponding fact for sets already in $\mathcal{A}$.

Lemma 1. For every $A \in \mathcal{A}$ and every $T \subseteq S$,

$$\mu^*(T) = \mu^*(T \cap A) + \mu^*(T \cap A^c). \tag{1}$$

Proof. Subadditivity gives

$$\mu^*(T) \leq \mu^*(T \cap A) + \mu^*(T \cap A^c).$$

For the reverse inequality, let

$$T \subseteq \bigcup_n B_n, \quad B_n \in \mathcal{A}.$$

Then

$$T \cap A \subseteq \bigcup_n (B_n \cap A), \quad T \cap A^c \subseteq \bigcup_n (B_n \cap A^c),$$

and hence

$$\begin{aligned} \mu^*(T \cap A) + \mu^*(T \cap A^c) &\leq \sum_n \mu(B_n \cap A) + \sum_n \mu(B_n \cap A^c) \\ &= \sum_n \mu(B_n). \end{aligned}$$

Taking the infimum over all such covers proves the claim. $\square$

Proposition 1. *For $E \subseteq S$, the following are equivalent.*

(i) *E is Carathéodory measurable.*

(ii) *For every $A \in \mathcal{A}$,*

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c). \quad (2)$$

(iii) *For every $T \subseteq S$,*

$$\mu^*(T) = \mu^*(T \cap E) + \mu^*(T \cap E^c). \quad (3)$$

Proof. If $\mu^*|_{\mathcal{A}[E]}$ is a premeasure, then for every $A \in \mathcal{A}$ the disjoint union

$$A = (A \cap E) \cup (A \cap E^c)$$

gives (2).

Suppose next that (2) holds, and let $T \subseteq S$. For any cover

$$T \subseteq \bigcup_n A_n, \quad A_n \in \mathcal{A},$$

we have

$$\begin{aligned} \mu^*(T \cap E) + \mu^*(T \cap E^c) &\leq \sum_n \mu^*(A_n \cap E) + \sum_n \mu^*(A_n \cap E^c) \\ &= \sum_n \mu^*(A_n) = \sum_n \mu(A_n). \end{aligned}$$

Taking the infimum over all such covers gives

$$\mu^*(T \cap E) + \mu^*(T \cap E^c) \leq \mu^*(T).$$

The reverse inequality is subadditivity, so (3) follows.

It remains to show that (3) implies measurability. Let $\mathcal{C}$ be the class of sets $F \subseteq S$ satisfying

$$\mu^*(T) = \mu^*(T \cap F) + \mu^*(T \cap F^c) \quad \text{for every } T \subseteq S.$$

By Lemma 1, $\mathcal{A} \subseteq \mathcal{C}$, and $\mathcal{C}$ is plainly closed under complements. It is also closed under finite unions. Indeed, if $F, G \in \mathcal{C}$, then

$$\begin{aligned}\mu^*(T) &= \mu^*(T \cap F) + \mu^*(T \cap F^c) \\ &= \mu^*(T \cap F) + \mu^*(T \cap F^c \cap G) + \mu^*(T \cap F^c \cap G^c) \\ &\geq \mu^*(T \cap (F \cup G)) + \mu^*(T \cap (F \cup G)^c),\end{aligned}$$

while the reverse inequality follows from subadditivity. Thus $\mathcal{C}$ is an algebra.

If E satisfies (3), then $E \in \mathcal{C}$, and therefore

$$\mathcal{A}[E] \subseteq \mathcal{C}.$$

Let $B_n \in \mathcal{A}[E]$ be pairwise disjoint and suppose $B = \bigcup_n B_n \in \mathcal{A}[E]$. Since every $B_n \in \mathcal{C}$, repeated use of (4), with $T = B$, gives for every N ,

$$\mu^*(B) = \sum_{n=1}^N \mu^*(B_n) + \mu^*\left(B \setminus \bigcup_{n=1}^N B_n\right).$$

Hence

$$\mu^*(B) \geq \sum_{n=1}^{\infty} \mu^*(B_n),$$

and countable subadditivity gives the reverse inequality. Thus $\mu^*|_{\mathcal{A}[E]}$ is a premeasure. $\square$

Condition (iii) is different from the definition we started with: it no longer refers to the algebra $\mathcal{A}$, but only to the fixed outer measure μ^* . This suggests considering all sets for which it holds at once. Let

$$\mathcal{M} = \{E \subseteq S : \mu^*(T) = \mu^*(T \cap E) + \mu^*(T \cap E^c) \text{ for every } T \subseteq S\}. \quad (4)$$

By Proposition 1, $\mathcal{M}$ is exactly the class of sets measurable in the sense defined above. We now ask what structure this class has. It contains $\mathcal{A}$; and, as we shall see, it is closed under complements and countable unions, while μ^* is countably additive on it. From here we arrive at Carathéodory's theorem by his classical argument.

Theorem 1 (Carathéodory). *The class $\mathcal{M}$ is a σ -algebra containing $\mathcal{A}$, and $\mu^*|_{\mathcal{M}}$ is a complete measure extending μ .*

Proof. We already know that $\mathcal{M}$ is an algebra containing $\mathcal{A}$. It remains to show closure under countable unions.

Let $E_n \in \mathcal{M}$ be pairwise disjoint and put

$$E = \bigcup_{n=1}^{\infty} E_n.$$

For every $T \subseteq S$, repeated use of (3) gives

$$\mu^*(T) = \sum_{n=1}^N \mu^*(T \cap E_n) + \mu^*\left(T \cap \left(\bigcup_{n=1}^N E_n\right)^c\right).$$

Since

$$T \cap E^c \subseteq T \cap \left(\bigcup_{n=1}^N E_n\right)^c,$$

we obtain

$$\mu^*(T) \geq \sum_{n=1}^N \mu^*(T \cap E_n) + \mu^*(T \cap E^c).$$

Letting $N \rightarrow \infty$ and using countable subadditivity on $T \cap E = \bigcup_n (T \cap E_n)$ gives

$$\mu^*(T) \geq \mu^*(T \cap E) + \mu^*(T \cap E^c).$$

The reverse inequality is subadditivity, so $E \in \mathcal{M}$.

For arbitrary $E_n \in \mathcal{M}$, replace them by the disjoint sequence

$$F_1 = E_1, \quad F_n = E_n \setminus \bigcup_{k < n} E_k.$$

Since $\mathcal{M}$ is an algebra, $F_n \in \mathcal{M}$, and the union is unchanged. Hence $\mathcal{M}$ is a σ -algebra.

Now let $E_n \in \mathcal{M}$ be pairwise disjoint and set $E = \bigcup_n E_n$. Since $E \in \mathcal{M}$, repeated use of (3), with $T = E$, gives

$$\mu^*(E) \geq \sum_{n=1}^{\infty} \mu^*(E_n),$$

while countable subadditivity gives the reverse inequality. Thus $\mu^*|_{\mathcal{M}}$ is a measure. It extends μ because $\mathcal{A} \subseteq \mathcal{M}$ and $\mu^* = \mu$ on $\mathcal{A}$.

Finally, let $N \in \mathcal{M}$ with $\mu^*(N) = 0$, and let $E \subseteq N$. Then $\mu^*(T \cap E) = 0$ for every $T \subseteq S$. Also $T \cap E^c \subseteq T$, while

$$T = (T \cap E) \cup (T \cap E^c),$$

so

$$\mu^*(T) \leq \mu^*(T \cap E) + \mu^*(T \cap E^c) = \mu^*(T \cap E^c) \leq \mu^*(T).$$

Hence E satisfies (3), and therefore $E \in \mathcal{M}$. The measure is complete. □